

1. Coding Skills



1



Next Section

State of the Art

Being a painter, Bruce thought of a new and futuristic idea for his artwork. He has **X** different paints on a palette and wants to create a painting with new paint combinations of the existing paints. He can utilize at most **Y** different paints for crafting a new paint combination.

Your task is to help Bruce find the number of different paint combinations he can craft from the existing paints on the palette and return an integer value representing the number of paints (new combinations and the existing paints).

Note:

- The volume of paint does not matter.
- Return answer modulo 10000.

Input Specification:

input1 : An integer value X, representing the number of different paints on the palette.

input2 : An integer value Y, representing the maximum number of different paints he can use to create a new paint combination.

Output Specification:

Return an integer value representing the number of paints (new combinations and the existing paints).

Example 1:

input1 : 5
input2 : 2

Attempted: 0/1

C

Compiler: gcc 5.4.0



```
1 #include<stdio.h>
2 #include<string.h>
3 // Read only region start
4
5 int numberOfPaints(int input1,int input2)
6 {
7     // Read only region end
8     // Write code here
9
10 }
```

☐ Use Custom Input

Compile and Test

Submit Code

3. Database Concept



1

2

3

4

5



Attempted: 2/5

Previous

Next

Question 3

Revisit Later

Assume that you are a developer working on general database concepts:

Rel1(LM->N)

Rel3(P->Q)

Rel2(N->P)

What are the minimal operations that can be performed to join relation Rel1 and Rel3 losslessly? Analyze the given choices and select the correct option.

Select an option

Clear Response

- ☐ Joining Rel1 and Rel3 directly
R(LMNPQ)
- ☒ Joining Rel1 to Rel2 and then Rel2 to Rel3
R(LMNPQ)
- ☐ Rel1 and Rel3 cannot be jointed together.
- ☐ It can not be determined.

3. Database Concept



1

2

3

4

5



Previous

Next

Attempted: 3/5

Question 4

Revisit Later

Assume that your company wants to keep track of its employees' information including names, addresses, and salaries. What is the most appropriate data model to store this information in a database management system? Select the appropriate answer from the given choices.

Select an option

Clear Response

☐ Object-oriented model☐ Network model☐ Hierarchical model☒ Relational model

3. Database Concept



1

2

3

4

5



Previous

Next Section

Attempted: 4/5

Question 5

Revisit Later

Consider that there is a deadlock situation in a bank's database management system which allows multiple customers to transfer money from one account to another at the same time. Which option denotes a probable reason for this scenario? Select the appropriate answer from the given choices.

Select an option

Clear Response

- ☐ A situation where one customer's transfer is blocked because the system is down
- ☐ A situation where one customer's transfer is blocked because they don't have enough money in their account
- ☒ A situation where two customers' transfers are blocked because they are waiting for each other to complete
- ☐ A situation where two customers try to transfer money from the same account at the same time

2. Data Structures



1

2

3

4

5

6



Previous Section

Next

Attempted: 0/6

Question 1

Revisit Later

Refer to the given singly linked list:

```
1 ListNode* f(ListNode* head){
2   ListNode* temp=head;
3   stack<ListNode*>st;
4   while(temp->next!=NULL){
5       st.push(temp);
6       temp=temp->next;
7   }
8   ListNode* list=create_list(st); //It only creates the list from the stack from
   top to bottom
9   return list;
10 }
```

If the function returns a list 1->2->3->4->5, what was the original list passed in the function? Analyze the given choices and select the correct option.

Select an option

Clear Response

☐ 5->4->3->2->1☒ 5->4->3->2->1->X (where X can be any value)☐ 1->2->3->4->5☐ X->5->4->3->2->1 (where X can be any value)

2. Data Structures



1

2

3

4

5

6



Previous

Next

Attempted: 1/6

Consider the given code that implements a stack using a linked list. What will be the output of this code? Analyze the given choices and select the correct option.

Select an option

Clear Response

☐ 2, 4, 6☒ 4, 2☐ 4, 6☐ 6

```
1 #include <iostream>
2 using namespace std;
3 struct Node {
4     int data;
5     Node* next;
6 };
7 class Stack {
8     private:
9     Node* top;
10    public:
11    Stack() { top = NULL; }
12    void push(int x) {
13        Node* temp = new Node;
14        temp->data = x;
15        temp->next = top;
16        top = temp;
17    }
18    void pop() {
19        if (top == NULL) return;
20        Node* temp = top;
21        top = top->next;
22        delete temp;
23    }
24    void display() {
25        Node* temp = top;
26        while (temp != NULL) {
27            cout << temp->data << " ";
28            temp = temp->next;
29        }
```

2. Data Structures



1

2

3

4

5

6



Previous

Next

Attempted: 2/6

Question 3



Revisit Later

Consider the given code:

```
1 priority_queue<int>pq;
2 int arr[] = {4, 3, 6, 2, 9, 1, 7, 7};
3 int ans = 0;
4 for (int i = 0; i < 8; i++) {
5     if (pq.size()) {
6         if (pq.top() > arr[i]) ans ^= arr[i];
7         else pq.push(arr[i]);
8     }
9     else pq.push(arr[i]);
10    cout << ans << " ";
11 }
```

What will be the output of the given code? Select the correct option from the given choices.

Select an option

Clear Response

☒ 0 3 3 1 1 0 7 0☐ 0 7 0 1 1 3 3 0☐ 0 3 3 1 1 0 0 0☐ 0 3 3 5 5 6 13 20

2. Data Structures



1

2

3

4

5

6



Previous

Next

Attempted: 3/6

Question 4

Revisit Later

Consider that you have the given array `arr[]={2,1,2,3,4,5,6,5,5,4}`.

If you insert all the elements in an integer set, what will be the size of the set? Select the correct option from the given choices.

Select an option

Clear Response

☐ 5.0☐ 10.0☐ 7.0☒ 6.0

2. Data Structures



1

2

3

4

5

6



Attempted: 4/5

Previous

Next

Question 5

Revisit Later

Assume that you are working on the given function which takes the root of a binary tree as a parameter and returns the sum of all the node values to the caller function.int:

```
1 fun(TreeNode* root)
2 {
3     if(root==NULL)return 0;
4     int l=fun(root->left);
5     int r=(root->right)->value;
6     return l+r+root->value;
7 }
```

However, the code is not giving the desired result. What should be the modified code? Analyze the given choices and select the correct option.

Select an option

Clear Response



```
1 int fun(TreeNode* root)
2 {
3     if(root==NULL)return 0;
4     int l=(root->left)->value;
5     int r=(root->right)->value;
6     return l+r+root->value;
7 }
```



```
1 int fun(TreeNode* root)
2 {
3     if(root==NULL)return 0;
4     int l=fun(root->left);
5     int r=fun(root->right);
6     return l + r + root->value;
7 }
```



```
1 int fun(TreeNode* root)
2 {
3     if(root==NULL)return 0;
4     int l=(root->left)->value;
5     int r=(root->right)->value;
6     return l->value + r->value + root->value;
7 }
```

2. Data Structures



1

2

3

4

5

6



Previous

Next Section

Attempted: 5/6

Question 6

Revisit Later

Consider an array `arr[] = {18, 32, 25, 41, 63, 11, 45}`. You want to find 18 using a binary search. What will be the first element you get when you start searching and what will be the last element before you get the desired element? Select the correct option from the given choices.

Select an option

Clear Response

☒ First element: 41, Last element: 32☐ First element: 32, Last element: 32☐ First element: 41, Last element: 18☐ First element: 25, Last element: 32

2. Data Structures



1

2

3

4

5

6



Previous

Next

Attempted: 1/6

Select an option

Clear Response

☐ 2, 4, 6☒ 4, 2☐ 4, 6☐ 6

```
11 Stack() { top = NULL; }
12 void push(int x) {
13     Node* temp = new Node;
14     temp->data = x;
15     temp->next = top;
16     top = temp;
17 }
18 void pop() {
19     if (top == NULL) return;
20     Node* temp = top;
21     top = top->next;
22     delete temp;
23 }
24 void display() {
25     Node* temp = top;
26     while (temp != NULL) {
27         cout << temp->data << " ";
28         temp = temp->next;
29     }
30     cout << endl;
31 }
32 };
33 int main() {
34     Stack s;
35     s.push(2);
36     s.push(4);
37     s.push(6);
38     s.pop();
39     s.display();
40     return 0;
41 }
```

4. Networking



1

2

3

4

5

6



Previous Section

Next

Attempted: 0/6

Question 1

Revisit Later

Assume that you are developing a web application that requires the user to input a password. Which is the most secure way to transmit the password from the user's browser to the server? Select the appropriate answer from the given choices.

Select an option

Clear Response

☐ Using HTTP☒ Using HTTPS with SSL/TLS encryption☐ Using FTP☐ Using Telnet

4. Networking



< 1 2 3 4 5 6 >



Previous

Next

Attempted: 1/6

Question 2

Revisit Later

Assume that you are working from home and want to access a file stored on a server in your office. Which layer determines the best path for your request for the file to travel and ensures that the request is delivered to the correct destination network? Select the correct answer from the given choices.



Select an option

Clear Response

☐ Transport layer☐ Data Link layer☒ Network layer☐ Physical layer

4. Networking



1

2

3

4

5

6



Previous

Next

Attempted: 2/6

Question 3

Revisit Later

Assume that a large file is being transmitted from a computer in New York to a computer in Los Angeles. The network administrator realizes that some data packets are not reaching the destination. What might be the possible reason? Select the appropriate answer from the given choices.

Select an option

Clear Response

- ☒ The size of the fragmented pack is 300 bytes
- ☐ The size of the fragmented pack is 200 bytes
- ☐ The size of the fragmented pack is 150 bytes
- ☐ The size of the fragmented pack is 250 bytes

4. Networking



1

2

3

4

5

6



Previous

Next

Attempted: 3/6

Question 4

Revisit Later

In your system, assume that an IP packet has arrived with the first 8 bits as 01101000. You need to identify the type of address used for better interpretation of the packet. According to you, what is the correct answer? Select the correct option from the given choices.



Select an option

Clear Response

☐ IPv7☒ IPv6☐ IPv5☐ IPv4

4. Networking



1

2

3

4

5

6



Previous

Next

Attempted: 4/5

Question 5

Revisit Later

Considering that you face a problem with the connection when you are trying to send an email, which layer of the OSI model may be the most likely source of the issue? Select the appropriate answer from the given choices.

Select an option

Clear Response

- ☒ Transport layer
- ☐ Network layer
- ☐ Data Link layer
- ☐ Physical layer

4. Networking



1

2

3

4

5

6



Previous

Next Section

Attempted: 5/6

Question 6

Revisit Later

Assume that you are transmitting a message consisting of 8-bit ASCII characters over a noisy channel using the checksum technique for error detection. The message is broken into 4-bit blocks and the checksum is calculated as the sum of all the 4-bit blocks. The message and checksum are transmitted, and the receiver calculates the checksum again. If the two checksums do not match, what can be concluded? Select the appropriate answer from the given choices.

Select an option

Clear Response

- ☐ The message has been received correctly
- ☒ The message has been received incorrectly
- ☐ The checksum has been calculated incorrectly
- ☐ The checksum has been transmitted incorrectly

5. OOPS



1

2

3

4

5

6

7



Previous Section

Next

Attempted: 0/7

Question 1

Revisit Later

As a software architect, assume that you are designing a new system that uses object-oriented programming principles. You want to ensure the system is flexible, scalable, and easy to maintain. In object-oriented programming, which *is not* a fundamental principle of the SOLID design principles? Select the correct answer from the given choices.

Interface Segregation Principle
Dependency Inversion Principle

Select an option

Clear Response

☐ Single Responsibility Principle☐ Open-Closed Principle☐ Liskov Substitution Principle☒ Program Execution Principle

5. OOPS



1

2

3

4

5

6

7



Previous

Next

Attempted: 1/7

Question 2

Revisit Later

Consider that you are a software developer building a vehicle management system for a transportation company. In order to represent different types of vehicles, you have a base class "Vehicle" with properties such as "make", "model", "year", and "price". Now, you want to create separate classes for cars, trucks, and motorcycles, each with specific properties. What is 'inheritance' in OOP? Select the correct option from the given choices.

Select an option

Clear Response

- ☐ A mechanism for code duplication and complexity
- ☒ A mechanism for code reuse and abstraction
- ☐ A mechanism for code encapsulation and modularity
- ☐ A mechanism for code isolation and security

5. OOPS



1

2

3

4

5

6

7



Previous

Next

Attempted: 1/7

Question 3

Revisit Later

Assume that you are creating a program to simulate a library system. The program should keep track of the books in the library, including the book's title, author, ISBN number, and availability. Each book should also have the ability to be checked out or returned. Considering the given requirements, which concept in Object-Oriented Programming best represents a library book in the program? Select the appropriate answer from the given choices.

Select an option

Clear Response



Object



Class



Inheritance



Polymorphism

5. OOPS



1

2

3

4

5

6

7



Previous

Next

Attempted: 2/7

Question 4

Revisit Later

Suppose you are designing a simple banking application using Object-Oriented Programming (OOPS) concepts.

Then, identify which of the following concepts would be best suited to represent a customer's bank account?

Select an option

Clear Response

☐ Class☒ Object☐ Inheritance☐ Polymorphism

5. OOPS



1

2

3

4

5

6

7



Previous

Next

Attempted: 3/7

Question 5

Revisit Later

Assume that a software developer is designing a code using the concept of class and object. He/she wants to create eight child classes named "Mercury", "Venus", "Earth", "Mars", "Jupiter", "Saturn", "Uranus", and "Neptune" that will inherit from a base class named "Planet". Which option should the developer use to design the base class? Select the correct answer from the given choices.

Select an option

Clear Response

☐ Inheritance☐ Encapsulation☒ Abstraction☐ Polymorphism

5. OOPS



1

2

3

4

5

6

7



Previous

Next

Attempted: 3/7

Question 6

Revisit Later

Assume that a software designer wants to create a class to represent an employee. The designer wants to ensure that the employee's personal information such as name, salary, and address can only be accessed and updated through methods provided by the class. Which OOP principle can the designer apply in this scenario? Select the correct option from the given choices.

Select an option

Clear Response

☐ Polymorphism☐ Inheritance☒ Encapsulation☐ Abstractionn

5. OOPS



1

2

3

4

5

6

7



Previous

Next Section

Attempted: 3/7

Question 7

Revisit Later

Considering that you are a developer, identify which code snippet demonstrates the concept of polymorphism in object-oriented programming. Analyze the given choices and select the correct option.



```
1 class Shape:
2     def area(self):
3         pass
4     class Rectangle(Shape):
5         def __init__(self, width, height):
6             self.width = width
7             &nbsp; &nbsp; &nbsp; &nbsp; self.height = height
8             &nbsp; &nbsp; &nbsp; def area(self):
9                 return self.width * self.height
10    class Circle(Shape):
11        def __init__(self, radius):
12            self.radius = radius
13            &nbsp; &nbsp; &nbsp; def area(self):
14                return 3.14 * self.radius ** 2
15    my_rectangle = Rectangle(4, 5)
16    my_circle = Circle(3)
17    print(my_rectangle.area())
18    print(my_circle.area())
```



```
1 class Vehicle:
2     def __init__(self, make, model):
3         self.make = make
4         &nbsp; &nbsp; &nbsp; &nbsp; self.model = model
5         &nbsp; &nbsp; &nbsp; def get_make_and_model(self):
6             return f"{self.make} {self.model}"
7     class Car(Vehicle):
8         def __init__(self, make, model, num_doors):
9             super().init(make, model)
```

5. OOPS



1

2

3

4

5

6

7



Previous

Next Section

Attempted: 3/7

Question 7

Revisit Later

Considering that you are a developer, identify which code snippet demonstrates the concept of polymorphism in object-oriented programming. Analyze the given choices and select the correct option.

```
18 print(my_circle.area())
```



```
1 class Vehicle:
2     def __init__(self, make, model):
3         self.make = make
4         &nbsp; &nbsp; &nbsp; &nbsp; &nbsp; self.model = model
5         &nbsp; &nbsp; &nbsp; def get_make_and_model(self):
6             return f"{self.make} {self.model}"
7         class Car(Vehicle):
8             def __init__(self, make, model, num_doors):
9                 super().__init__(make, model)
10                self.num_doors = num_doors
11        class Truck(Vehicle):
12            def __init__(self, make, model, payload_capacity):
13                super().__init__(make, model)
14                self.payload_capacity = payload_capacity
15        my_car = Car("Toyota", "Corolla", 4)
16        my_truck = Truck("Ford", "F-150", 2000)
17        print(my_car.get_make_and_model())
18        print(my_truck.get_make_and_model())
```



```
1 class Employee:
2     def __init__(self, name, salary):
3         self.name = name
4         &nbsp; &nbsp; &nbsp; &nbsp; &nbsp; self.salary = salary
5         &nbsp; &nbsp; &nbsp; def calculate_pay(self, hours_worked):
6             return self.salary * hours_worked
7         class Manager(Employee):
```

5. OOPS



1

2

3

4

5

6

7



Previous

Next Section

Attempted: 3/7

Question 7

Revisit Later

Considering that you are a developer, identify which code snippet demonstrates the concept of polymorphism in object-oriented programming. Analyze the given choices and select the correct option.

```
11 class Truck(Vehicle):
12 def __init__(self, make, model, payload_capacity):
13 super().__init__(make, model)
14 self.payload_capacity = payload_capacity
15 my_car = Car("Toyota", "Corolla", 4)
16 my_truck = Truck("Ford", "F-150", 2000)
17 print(my_car.get_make_and_model())
18 print(my_truck.get_make_and_model())
```



```
1 class Employee:
2 def __init__(self, name, salary):
3 self.name = name
4 &nbsp; &nbsp; &nbsp; &nbsp; &nbsp; self.salary = salary
5 &nbsp; &nbsp; &nbsp; def calculate_pay(self, hours_worked):
6 return self.salary * hours_worked
7 class Manager(Employee):
8 def __init__(self, name, salary, bonus_rate):
9 super().__init__(name, salary)
10 self.bonus_rate = bonus_rate
11 &nbsp; &nbsp; &nbsp; def calculate_pay(self, hours_worked):
12 base_pay = super().calculate_pay(hours_worked)
13 return base_pay + self.bonus_rate * base_pay
14 my_employee = Employee("Alice", 10)
15 my_manager = Manager("Bob", 20, 0.2)
16 print(my_employee.calculate_pay(40))
17 print(my_manager.calculate_pay(40))
```

6. Operating System



1

2

3

4

5

6



Previous Section

Next

Attempted: 0/6

Question 1

Revisit Later

As a software developer, assume that you are developing an application that requires a large amount of memory to run. You are concerned that the application may run out of memory and crash. Which memory management technique will be the best solution to this problem? Select the correct answer from the given choices.

Select an option

Clear Response

☐ Segmentation☐ Paging☒ Virtual Memory☐ Memory Compression

6. Operating System



1

2

3

4

5

6



Previous

Next

Attempted: 1/6

Question 3

Revisit Later

Assume that four processes, A, B, C, and D, share a database resource. Process A holds a read lock on a table and waits for a write lock. Process B holds a write lock on the table and waits for a read lock. Process C holds a read lock on another table and waits for the write lock held by Process B, while Process D holds a write lock on yet another table and waits for the read lock held by Process A. The given scenario denotes an example of which situation? Select the correct answer from the given choices.

Select an option

Clear Response

☐ Time slice exhaustion☐ Livelock☐ Resource depletion☒ Circular wait

6. Operating System



1

2

3

4

5

6



Previous

Next

Attempted: 1/6

Question 4

Revisit Later

Assume that you are working on a project in which you have to support an array of behavior, including timesharing multiple processes so that various tasks are available. Which type of management will you choose to accomplish this task? Select the correct answer from the given choices.

Select an option

Clear Response

☒ Application management☐ Memory management☐ Time stamping management☐ Disk management

6. Operating System



1

2

3

4

5

6



Previous

Next

Attempted: 1/6

Question 5

Revisit Later

Assume that you work as a software developer and you need to run a large program that requires more memory than your computer has available. Which option indicates a statement that is not a benefit of virtual memory in this situation? Select the appropriate answer from the given choices.

Select an option

Clear Response

- ☐ Allows for efficient use of physical memory
- ☒ Decreases disk I/O by eliminating the need for a swap file
- ☐ Enables the large program to be run despite limited physical memory
- ☐ Provides a mechanism for memory protection and security

6. Operating System



1

2

3

4

5

6



Previous

Attempted: 1/6

Question 6

Revisit Later

Consider that you are giving a job interview in which the interviewer asks you to find out the output of the given code snippet:

```
1 #include <stdio.h>
2 #include <unistd.h>
3 int main() {
4     fork();
5     fork();
6     fork();
7     printf("fork ");
8     return 0;
9 }
```

What will be the output of the given above? Select the correct answer from the given choices.

Select an option

Clear Response

☐ fork fork fork fork fork fork*8 fork*☐ fork fork fork☐ fork☐ 0.0

6. Operating System



1

2

3

4

5

6



Previous

Next

Attempted: 3/6

Question 3

Revisit Later

Assume that four processes, A, B, C, and D, share a database resource. Process A holds a read lock on a table and waits for a write lock. Process B holds a write lock on the table and waits for a read lock. Process C holds a read lock on another table and waits for the write lock held by Process B, while Process D holds a write lock on yet another table and waits for the read lock held by Process A. The given scenario denotes an example of which situation? Select the correct answer from the given choices.

Select an option

Clear Response

☐ Time slice exhaustion☐ Livelock☐ Resource depletion☒ Circular wait

5. OOPS



1

2

3

4

5

6

7



Previous

Next Section

Attempted: 3/7

Question 7

Revisit Later

Considering that you are a developer, identify which code snippet demonstrates the concept of polymorphism in object-oriented programming. Analyze the given choices and select the correct option.

☐

```
1 class Shape:
2     def area(self):
3         pass
4 class Rectangle(Shape):
5     def __init__(self, width, height):
6         self.width = width
7                     self.height = height
8                 def area(self):
9             return self.width * self.height
10 class Circle(Shape):
11     def __init__(self, radius):
12         self.radius = radius
13                 def area(self):
14             return 3.14 * self.radius ** 2
15 my_rectangle = Rectangle(4, 5)
16 my_circle = Circle(3)
17 print(my_rectangle.area())
18 print(my_circle.area())
```

☐

```
1 class Vehicle:
2     def __init__(self, make, model):
3         self.make = make
4                     self.model = model
```

5. OOPS



1

2

3

4

5

6

7



Previous

Next Section

Question 7

Revisit Later

Considering that you are a developer, identify which code snippet demonstrates the concept of polymorphism in object-oriented programming. Analyze the given choices and select the correct option.

Attempted: 3/7

```
17 print(my_rectangle.area())
18 print(my_circle.area())
```

☐

```
1 class Vehicle:
2     def __init__(self, make, model):
3         self.make = make
4         &nbsp; &nbsp; &nbsp; &nbsp; &nbsp; self.model = model
5         &nbsp; &nbsp; &nbsp; def get_make_and_model(self):
6             return f"{self.make} {self.model}"
7     class Car(Vehicle):
8     def __init__(self, make, model, num_doors):
9         super().__init__(make, model)
10        self.num_doors = num_doors
11    class Truck(Vehicle):
12    def __init__(self, make, model, payload_capacity):
13        super().__init__(make, model)
14        self.payload_capacity = payload_capacity
15    my_car = Car("Toyota", "Corolla", 4)
16    my_truck = Truck("Ford", "F-150", 2000)
17    print(my_car.get_make_and_model())
18    print(my_truck.get_make_and_model())
```

☐

```
1 class Employee:
```

5. OOPS



1

2

3

4

5

6

7



Previous

Next Section

Question 7

Revisit Later

Considering that you are a developer, identify which code snippet demonstrates the concept of polymorphism in object-oriented programming. Analyze the given choices and select the correct option.

Attempted: 3/7

```
15 my_car = Car("Toyota", "Corolla", 4)
16 my_truck = Truck("Ford", "F-150", 2000)
17 print(my_car.get_make_and_model())
18 print(my_truck.get_make_and_model())
```

☐

```
1 class Employee:
2     def __init__(self, name, salary):
3         self.name = name
4         &nbsp; &nbsp; &nbsp; &nbsp; self.salary = salary
5         &nbsp; &nbsp; &nbsp; def calculate_pay(self, hours_worked):
6             return self.salary * hours_worked
7     class Manager(Employee):
8         def __init__(self, name, salary, bonus_rate):
9             super().__init__(name, salary)
10            self.bonus_rate = bonus_rate
11            &nbsp; &nbsp; &nbsp; def calculate_pay(self, hours_worked):
12                base_pay = super().calculate_pay(hours_worked)
13                return base_pay + self.bonus_rate * base_pay
14 my_employee = Employee("Alice", 10)
15 my_manager = Manager("Bob", 20, 0.2)
16 print(my_employee.calculate_pay(40))
17 print(my_manager.calculate_pay(40))
```

5. OOPS



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Previous

Next

Attempted: 3/7

Question 6

Revisit Later

Assume that a software designer wants to create a class to represent an employee. The designer wants to ensure that the employee's personal information such as name, salary, and address can only be accessed and updated through methods provided by the class. Which OOP principle can the designer apply in this scenario? Select the correct option from the given choices.

Select an option

Clear Response

☐ Polymorphism☐ Inheritance☐ Encapsulation☐ Abstractionn

5. OOPS



1

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6

7



Previous

Next Section

Attempted: 3/7

Question 7

Revisit Later

Select an option

Clear Response

Considering that you are a developer, identify which code snippet demonstrates the concept of polymorphism in object-oriented programming. Analyze the given choices and select the correct option.

☐

```
1 class Animal:
2 def speak(self):
3 pass
4 class Dog(Animal):
5 def speak(self):
6 return "Woof!"
7 class Cat(Animal):
8 def speak(self):
9 return "Meow!"
10 def pet_speak(pet):
11 print(pet.speak())
12 my_dog = Dog()
13 my_cat = Cat()
14 pet_speak(my_dog)
15 pet_speak(my_cat)
```

☐

```
1 class Shape:
2 def area(self):
3 pass
4 class Rectangle(Shape):
5 def init (self, width, height):
```

5. OOPS



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7



Previous

Next

Attempted: 5/7

Question 4

Revisit Later

Suppose you are designing a simple banking application using Object-Oriented Programming (OOPS) concepts.

Then, identify which of the following concepts would be best suited to represent a customer's bank account?

Select an option

Clear Response



Class



Object



Inheritance



Polymorphism





5. OOPS



1

2

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4

5

6

7



Previous

Next

Attempted: 4/7

Question 5

Revisit Later

Assume that a software developer is designing a code using the concept of class and object. He/she wants to create eight child classes named "Mercury", "Venus", "Earth", "Mars", "Jupiter", "Saturn", "Uranus", and "Neptune" that will inherit from a base class named "Planet". Which option should the developer use to design the base class? Select the correct answer from the given choices.

Select an option

Clear Response

☐

Inheritance

☐

Encapsulation

☐

Abstraction

☐

Polymorphism





5. OOPS



1

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4

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Previous

Next

Attempted: 6/7

Question 4

Revisit Later

Suppose you are designing a simple banking application using Object-Oriented Programming (OOPS) concepts.

Then, identify which of the following concepts would be best suited to represent a customer's bank account?

Select an option

Clear Response



Class



Object



Inheritance



Polymorphism





5. OOPS



1

2

3

4

5

6

7



Previous Section

Next

Attempted: 6/7

Question 1

Revisit Later

As a software architect, assume that you are designing a new system that uses object-oriented programming principles. You want to ensure the system is flexible, scalable, and easy to maintain. In object-oriented programming, which *is not* a fundamental principle of the SOLID design principles? Select the correct answer from the given choices.

Select an option

Clear Response

☐ Single Responsibility Principle☐ Open-Closed Principle☐ Liskov Substitution Principle☐ Program Execution Principle

5. OOPS



1

2

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4

5

6

7



Previous

Next

Attempted: 6/7

Question 3

Revisit Later

Assume that you are creating a program to simulate a library system. The program should keep track of the books in the library, including the book's title, author, ISBN number, and availability. Each book should also have the ability to be checked out or returned. Considering the given requirements, which concept in Object-Oriented Programming best represents a library book in the program? Select the appropriate answer from the given choices.

Select an option

Clear Response

☐ Object☒ Class☐ Inheritance☐ Polymorphism

2. Data Structures



1

2

3

4

5

6



Previous

Next

Attempted: 6/6

Question 3

Revisit Later

Select an option

Clear Response

Consider the given code:

```
1 priority_queue<int>pq;
2 int arr[] = {4, 3, 6, 2, 9, 1, 7, 7};
3 int ans = 0;
4 for (int i = 0; i < 8; i++) {
5     if (pq.size()) {
6         if (pq.top() > arr[i]) ans ^= arr[i];
7         else pq.push(arr[i]);
8     }
9     else pq.push(arr[i]);
10    cout << ans << " ";
11 }
```

What will be the output of the given code? Select the correct option from the given choices.

☐ 0 3 3 1 1 0 7 0☐ 0 7 0 1 1 3 3 0☒ 0 3 3 1 1 0 0 0☐ 0 3 3 5 5 6 13 20